```
lcd_msg("ACCESS", "DENIED")

print("Unauthorized RFID")

except KeyboardInterrupt:

GPIO.cleanup()

Break
```

7. Result:

PRACTICAL – 16

1. Aim

Create a motion detector by interfacing camera and 16X2 display with Raspberry Pi, Print the appropriate message on the display.

2. Prior Concepts

i) Concept of Python Programming.

ii) Basics of camera operation.

iii) Basics of 12C communication.

iv) Interfacing 16x2 LCD with Raspberry Pi.

v) Basic image processing concepts.

3. New Concepts

i) Motion detection using computer vision.

ii) Raspberry Pi Camera interfacing.

iii) Frame differencing technique.

iv) Integration of camera and LCD modules.

4. Hardware Required

i) Raspberry Pi Camera Module V2/V3

ii) 16×2 LCD with 12C (PCF8574)

iii) Jumper Wires

5. Procedure

Pin Connection

Camera Module: Connect the Raspberry Pi Camera Module to the CSI Camera Port of Raspberry Pi 4.

12C LCD Connections to Raspi:

LCD (12C)	Raspberry Pi 4
VCC	5V (Pin 2)
GND	GND (Pin 6)
SDA	GPIO2 SDA (Pin 3)
SCL	GPIO3 SCL (Pin 5)

Input IMAGE

1) Install Python3 on Raspi.

Write the below code in command prompt:

sudo apt install idle3

Enable following Interfaces:

sudo raspi-config

12C -> Enable

Camera -> Enable

sudo reboot

Install Libraries

sudo apt update

sudo apt install python3-opencv -y

pip3 install RPLCD

pip3 install smbus2

pip3 install picamera2

6. Implementation with Python Code

Code:

```
from picamera2 import Picamera?

import cv.2

import numpy as np

import time

from RPLCD.i2c import CharLCD

Icd = CharLCD(

PCF8574',
```

```
0x27,
cols=16,
rows 2
)
pcam2 Picamera2()
config = picam2.create_preview_configuration(
main=["size": (640,480))
pram2.configure(config)
picam2.start()
time.sleep(2)
frame1 = picam2.capture_array()
frame1 = cv2.cvtColor(frame1, cv2.COLOR_BGR2GRAY)
while True:
frame2 = picam2.capture_array()
frame2_gray = cv2.cvtColor(
frame2,
cv2.COLOR_BGR2GRAY
)
diff = cv2.absdiff(
frame1,
```

```
frame2_gray

)

threshold= cv2.threshold(

diff,

25,

255,

cv2.THRESH_BINARY

) [1]

motion_pixels = cv2.countNonZero(

threshold

)

lcd.clear()

if motion_pixels > 5000:

print("Motion Detected")

lcd.write_string(

"Motion Detected"

)

else:

print("No Motion")

lcd.write_string(
```

"No Motion"

)

frame1 = frame2_gray

time.sleep(1)

7. Result:

PRACTICAL-17

1. Aim

push-button and an LED with a web dashboard. The flow must allow users to toggle the LED from the dashboard and via the physical button. A button press should force the LED ON Design and implement a Node-RED flow on Raspberry Pi that integrates a physic for 5 seconds and then automatically restore the LED to the dashboard's last state.

2. Prior Concepts

i) Concept of Python Programming

ii) LED interfacing with Raspberry Pi

iii) Push button switch operation

iv) Web browser operation

v) Event-driven programming basics

3. New Concepts

i) Node-RED visual programming.

ii) Flow-based loT application development.